Hermite-rank prediction and fractional-time obstructions under inverse-number smoothing

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Abstract

For a square-integrable function of one standard Gaussian coordinate, conditioning on any Gaussian observation family transports its Hermite expansion to the orthogonal projection of that coordinate. We reconstruct the full L^2 prediction formula, exact prediction-energy series and a quantitative leading-rank remainder. Applied to the actual full past of fractional Brownian motion, these identities give explicit lower bounds with exponent $2m\min(H, 1-H)$ for any nonzero Hermite coefficient of rank m, and exclude higher-power increment bounds at zero information. The inverse-number potential divides positive-rank coefficients by their rank, so it preserves this obstruction despite being compact on a single Gaussian observation space. A concrete quarter-Hurst covariance witness also separates full-past prediction from current-value prediction. The projection and Hermite mechanisms are classical, and the two fractional covariance regimes are already known in primary literature. This paper supplies complete proofs and exact frozen formal correspondences; no priority claim is made.

1 Setting and main prediction formula

Let $X : \mathcal{H} \to L^2(\Omega, P)$ be an isonormal real Gaussian family on a real separable Hilbert space: X is linear, centered, jointly Gaussian, and $\mathbb{E}[X(h)X(k)] = \langle h, k\rangle$. Fix $u \in \mathcal{H}$ with $\|u\| = 1$. Let $(h_i)_{i\in I}$ be any observation family, and put

$$M = \overline{\operatorname{span}}\{h_i : i \in I\}, \quad p = P_M u, \quad v = \|p\|^2 \in [0,1], \quad \mathcal{A} = \sigma(X(h_i) : i \in I).$$

We use L^2 equivalence classes; observation sigma-algebras are not identified literally when they differ on null sets.

Define Wick polynomials $W_n(h)$ by the generating identity

$$\exp(tX(h) - t^2\|h\|^2/2) = \sum_{n=0}^{\infty} \frac{t^n}{n!} W_n(h). \tag{1}$$

Thus $W_0(h) = 1$, $W_1(h) = X(h)$ and $W_2(h) = X(h)^2 - \|h\|^2$. At unit u, these are the probabilists' Hermite polynomials $\mathrm{He}_n(X(u))$.

Theorem 1.1. *Every $f \in L^2(\sigma(X(u)), P)$ has a unique expansion*

$$f = \sum_{n\geq 0} a_n W_n(u), \qquad a_n = \frac{\mathbb{E}[fW_n(u)]}{n!}, \qquad \|f\|_2^2 = \sum_{n\geq 0} a_n^2 n!. \tag{2}$$

Conditioning gives the norm-convergent series and exact energy

$$\mathbb{E}[f \mid \mathcal{A}] = \sum_{n\geq 0} a_n W_n(p), \qquad \|\mathbb{E}[f \mid \mathcal{A}]\|_2^2 = \sum_{n\geq 0} a_n^2 n! v^n. \tag{3}$$

If $a_n = 0$ for $n < m$, and $A_m = a_m^2 m!$, then

$$A_m v^m \leq \|\mathbb{E}[f \mid \mathcal{A}]\|_2^2 \leq v^m \|f\|_2^2, \tag{4}$$

$$0 \leq \|\mathbb{E}[f \mid \mathcal{A}]\|_2^2 - A_m v^m \leq v^{m+1} \|f\|_2^2. \tag{5}$$

For any varying observation families with $v > 0$ and $v \to 0$,

$$\frac{\|\mathbb{E}[f \mid \mathcal{A}]\|_2^2}{v^m} \longrightarrow A_m, \qquad 0 \leq \frac{\|\mathbb{E}[f \mid \mathcal{A}]\|_2^2}{v^m} - A_m \leq v\|f\|_2^2. \tag{6}$$

The exact nonlinear prediction mechanism is established Gaussian theory. McElroy and Das [1] give Hermite-based nonlinear prediction formulas for Gaussian transformations with finite data. Here the Hilbert projection describes arbitrary observation families and makes the L^2 completion and rank remainder explicit. This is a formalization exposition, not a claim to have discovered nonlinear Gaussian prediction.

2 Completeness and the conditional Wick identity

Lemma 2.1. *For all $h, k \in \mathcal{H}$ and nonnegative integers n, m,*

$$\mathbb{E}[W_n(h) W_m(k)] = \mathbf{1}_{n=m}\, n! \langle h, k\rangle^n.$$

The polynomials $W_n(u)$ have dense linear span in $L^2(\sigma(X(u)), P)$.

Proof. The Gaussian exponential formula gives

$$\mathbb{E}\left[e^{sX(h) - s^2\|h\|^2/2} e^{tX(k) - t^2\|k\|^2/2}\right] = e^{st\langle h,k\rangle}.$$

Taking coefficients yields orthogonality and its exact constant. Coefficient extraction under expectation is valid: on any compact set of real or complex parameters, a Gaussian exponential with a fixed larger real linear coefficient dominates each differentiated expression in L^1, by the Gaussian exponential moment formula.

To prove completeness rather than assume it, identify the terminal space with $L^2(\gamma)$, γ standard Gaussian measure. If g is orthogonal to all polynomials, consider $F(z) = \int g(x) e^{zx}\, d\gamma(x)$. Cauchy–Schwarz and Gaussian exponential moments make this function entire, with derivatives obtained under the integral uniformly on compact parameter sets. All its Taylor coefficients vanish, so F is identically zero. In particular the finite signed measure $g\gamma$ has zero Fourier transform. For completeness, Fourier uniqueness here follows by Gaussian convolution: Fourier inversion for a Gaussian and Fubini show each Gaussian convolution of this measure is zero; taking an approximate identity gives zero integrals against every compactly supported continuous function, and hence the measure is zero. Thus $g = 0$. The Wick polynomials span all polynomials, because their leading coefficients are one. The orthogonal complement is therefore zero, proving density and, by Hilbert-space orthogonal expansion, (2). □

Lemma 2.2. $\mathbb{E}[W_n(u) \mid \mathcal{A}] = W_n(p)$ *for every n.*

Proof. Since $u = p + (u - p)$ and $u - p \perp M$, $X(u - p)$ is independent of every finite vector of observations: their joint Gaussian characteristic function factors because all cross-covariances are zero. A pi-lambda argument extends independence to $\mathcal{A}$. Also $X(p)$ is $\mathcal{A}$-measurable modulo null sets. Indeed p is a norm limit of finite linear combinations of observation directions, so $X(p)$ is an L^2 limit of measurable variables and has a measurable representative. Consequently

$$\mathbb{E}[e^{tX(u)-t^2/2} \mid \mathcal{A}] = e^{tX(p)-t^2\|p\|^2/2}.$$

The residual exponential has expectation one, giving the equality. The exponential series is analytic in L^2 on compact parameter sets: the same Gaussian domination as in Lemma 2.1, now using squared norms, justifies differentiating. Conditional expectation is a bounded linear map on L^2, so differentiation commutes with it. Taking the nth derivative at zero proves the claim. □

Proof of Theorem 1.1. Lemma 2.1 gives the terminal expansion and Parseval. Conditional expectation is an L^2 contraction, so it commutes with its partial-sum limit. Lemma 2.2 yields the first formula in (3). The terms are mutually orthogonal and have squared norms $a_n^2 n! v^n$; taking norms of finite sums and passing to the limit gives the energy formula. If the coefficients below m vanish, all terms start at m. Since $0 \le v \le 1$, their sum is at most $v^m \sum_n a_n^2 n!$, and at least its mth term. After subtracting that term, every remaining power is at most v^{m+1}, proving (5). For $v > 0$ division by v^m proves the last bound in (6); the squeeze theorem proves its limit. □

If $v = 0$, the predictor is $a_0 = \mathbb{E}f$, with $0^0 = 1$ in the series. No normalized-rank quotient is then used. If f is centered and nonzero, Parseval ensures a least positive m with $a_m \neq 0$; this is its Hermite rank. The upper bound in (4) is attained for a pure rank-m polynomial. No assumption that f itself is polynomial is needed anywhere.

3 Actual fractional pasts and the two covariance regimes

Let B^H be a continuous fractional Brownian motion with $0 < H < 1$, normalized by

$$R_H(s,t) = \mathbb{E}[B_s^H B_t^H] = \tfrac{1}{2}(s^{2H} + t^{2H} - |t-s|^{2H}), \qquad s,t \ge 0.$$

It has the Gaussian Hilbert realization $B_t^H = X(u_t)$, $\langle u_s, u_t \rangle = R_H(s,t)$. In particular $\|u_1\| = 1$. For $0 \le t \le 1$ put

$$\mathcal{F}_t^H = \sigma(B_s^H : 0 \le s \le t), \quad P_t = \mathbb{E}[\,\cdot \mid \mathcal{F}_t^H], \quad v_H(t) = \|P_t B_1^H\|_2^2, \quad \kappa_H = 2\min(H, 1-H).$$

These are conditional expectations onto the entire actual observed past. In the frozen construction a continuous modification realizes the Hilbert coordinates. Coordinatewise almost-sure equality suffices to identify their L^2 observation spaces: the sets agreeing modulo null sets form a sigma-algebra containing each coordinate preimage. Closure under countable unions only uses countable unions of null sets. Thus the augmented observation sigma-algebras agree, and the conditional expectations agree as L^2 classes. No common exceptional set for an uncountable family or literal equality of raw sigma-algebras is assumed.

Proposition 3.1. *For $0 < t \le 1$,*

$$v_H(t) \ge \tfrac{1}{4} t^{\kappa_H}. \tag{7}$$

Consequently, for $f \in L^2(\sigma(B_1^H), P)$ with terminal Hermite coefficients a_n and every nonnegative integer m,

$$\|P_t f\|_2^2 \ge \frac{a_m^2 m!}{4^m} t^{\kappa_H m}. \tag{8}$$

If f is centered, $a_m \neq 0$, and $\alpha > \kappa_H m$, then for every real C and every $\delta > 0$ some $0 < t < \min(\delta, 1)$ satisfies

$$Ct^\alpha < \|P_t f - P_0 f\|_2^2. \tag{9}$$

Proof. The current observation is one member of the past. Orthogonality of conditional projection therefore gives

$$\langle P_t B_1^H, B_t^H \rangle = R_H(1,t), \qquad v_H(t) \geq R_H(1,t)^2 / t^{2H}.$$

Write $r = R_H(1,t) = (1 + t^{2H} - (1-t)^{2H})/2$. Since $(1-t)^{2H} \leq 1$, $r \geq t^{2H}/2$. Also $r \geq t/2$: if $2H \leq 1$ use $t^{2H} \geq t$ in the preceding bound, while if $2H \geq 1$ use $(1-t)^{2H} \leq 1 - t$ in the exact formula. The two bounds give

$$v_H(t) \geq \tfrac{1}{4} \max(t^{2H}, t^{2-2H}) = \tfrac{1}{4} t^{2\min(H, 1-H)}.$$

Theorem 1.1 and the single nonnegative rank-m energy term now prove (8).

The zero-time past is generated by $B_0^H = 0$, so for centered f, $P_0 f = 0$. Put $K = a_m^2 m!/4^m > 0$ and $\beta = \kappa_H m$. Since $\alpha > \beta$, $Ct^{\alpha-\beta} \to 0$ as $t \downarrow 0$. Choose t smaller than δ and one, with this expression less than K. Multiplying by t^β and using (8) gives (9). □

The lower constant $1/4$ is not claimed optimal, and this proof does not give an asymptotic equivalence for v_H. Sottinen and Viitasaari's primary prediction-law paper [2] gives explicit full-past conditional laws and no-information covariance asymptotics in the two regimes. Our elementary lower bound uses only the covariance and one current observation to obstruct any faster full-past decay. It does not claim discovery of those exponents or replace the explicit prediction kernel.

4 Inverse-number smoothing preserves the obstruction

Define on the one-coordinate terminal space the inverse-number potential

$$\mathcal{U} f = \sum_{n \geq 1} \frac{a_n}{n} W_n(u), \qquad f = \sum_{n \geq 0} a_n W_n(u). \tag{10}$$

It removes the mean. In the full Gaussian library this is the bounded inverse-number operator constructed as the divergence recovery map followed by its Hilbert adjoint; its action on each positive chaos is $1/n$. The diagonal formula (10) is its exact restriction, not a different regularizer.

Proposition 4.1. *$\mathcal{U}$ is bounded, self-adjoint, nonnegative and compact on $L^2(\sigma(X(u)), P)$. Each positive coefficient is nonzero after smoothing exactly when it was nonzero before smoothing; in particular the least positive Hermite rank is preserved. Moreover*

$$\|\mathbb{E}[\mathcal{U} f \mid \mathcal{A}]\|_2^2 = \sum_{n \geq 1} \frac{a_n^2}{n^2} n! v^n. \tag{11}$$

For fractional terminal f and any $m \geq 1$ with $a_m \neq 0$,

$$\|P_t \mathcal{U} f\|_2^2 \geq \frac{a_m^2 m!}{m^2 4^m} t^{\kappa_H m}. \tag{12}$$

Every higher-power increment bound as in (9) therefore fails for $\mathcal{U} f$ as well. Centering of the original f is unnecessary.

Proof. In the orthonormal basis $W_n(u)/\sqrt{n!}$, $\mathcal{U}$ is diagonal with eigenvalues $0, 1, 1/2, 1/3, \ldots$. These real nonnegative multipliers prove boundedness, self-adjointness and nonnegativity. Truncating after rank N gives a finite-rank operator with norm error $1/(N+1)$; hence $\mathcal{U}$ is compact. Its positive coefficients are a_n/n, so positive ranks are preserved. Apply Theorem 1.1 to this new terminal function to obtain (11), and Proposition 3.1 to get (12). The latter has positive constant and $P_0\mathcal{U}f = 0$; the same elementary power comparison proves the increment obstruction. □

This compactness is on a *single* Gaussian observation space. It does not imply compactness of the full Gaussian potential on an infinite-dimensional source: on first chaos the potential acts as the identity, and an infinite orthonormal family has no norm-convergent subsequence. Nor does compact smoothing in Hermite rank imply better physical-time regularity of conditional forecasts. The obstruction retains its positive-rank power because the corresponding coefficient is divided by a fixed nonzero integer.

5 A concrete full-past versus current-value witness

The Hilbert projection in the prediction formula must use the correct observation family. An explicit example already distinguishes the full past from the present coordinate.

Proposition 5.1. *Let $H = 1/4$ and define*

$$F = \mathbb{E}[B_4^H \mid \mathcal{F}_2^H], \qquad C = \mathbb{E}[B_4^H \mid \sigma(B_2^H)].$$

Then $F \neq C$ in L^2, and $\|F\|_2^2 > \|C\|_2^2$.

Proof. The exact covariance formula gives

$$R_H(4,2) = 1, \quad R_H(4,1) = \tfrac{1}{2}(3-\sqrt{3}), \quad R_H(2,1) = \tfrac{1}{2}\sqrt{2}, \quad R_H(2,2) = \sqrt{2}.$$

The one-observation projection is $C = B_2^H/\sqrt{2}$, so $\langle C, B_1^H\rangle = 1/2$. Since B_1^H is in the full past, $\langle F, B_1^H\rangle = R_H(4,1) = (3-\sqrt{3})/2$, a different number. Thus $F \neq C$. The one-observation subspace is contained in the full-past L^2 subspace. Both are orthogonal projections of B_4^H; their projection identities give $F - C \perp C$. Pythagoras yields $\|F\|_2^2 = \|C\|_2^2 + \|F-C\|_2^2 > \|C\|_2^2$. □

This witness does not assert that fractional Brownian motion is a new non-Markovian example. It records exactly why replacing a full-past projection with the current-coordinate projection changes the theorem.

6 Formal scope and primary comparisons

The verified baseline is `bab688bb6140cd6316c80680095a914ea0d346db`. The core correspondence is `GaussianWickConditionalProjection`, `GaussianTerminalPredictionSpectrum` and `GaussianPredictionRankAsymptotic`. Actual continuous-past identification is in `ContinuousFractionalPrediction`; the two-regime obstruction is `FractionalDualRegimePrediction`. The inverse-number application is `GaussianPotentialPredictionRank`, with the actual operator in `GaussianInverseNumberPotential`. The quarter-Hurst witness is `FractionalCurrentObservationObstruction`. Exact declarations, original hypotheses, source hashes and human proof reorganizations accompany the manuscript.

The elementary noncompactness of the full potential is a classical deduction from its exact first-chaos identity, not a newly named formal theorem.
McElroy–Das's primary introduction describes the Hermite predictor and mean-square-error mechanism for finite data; access to its complete main-theorem text was incomplete, so no detailed theorem-number priority comparison is claimed. Sottinen–Viitasaari's accessible primary paper, Theorem 3.1 and Proposition 3.2, gives the conditional law and the two small-past covariance asymptotics. Our proof retains a weaker explicit uniform lower constant and does not assert a new fractional exponent. The methods and central structure are classical. This is a consolidated formalization companion with explicit domain and regularity diagnostics; novelty remains unresolved. It neither completes the paused Bakry/fBM campaign nor proves new curvature, temporal stochastic integration or neural-network performance claims. Full baseline verification evidence was hash-checked without repeating a research build.

References

[1] T. McElroy and S. Das, *Nonlinear prediction via Hermite transformation*, Statistical Theory and Related Fields 5 (2021), 49–54. doi:10.1080/24754269.2020.1856589.

[2] T. Sottinen and L. Viitasaari, *Prediction law of fractional Brownian motion*, Statist. Probab. Lett. 129 (2017), 155–166. arXiv:1610.02900.